Review #5

1 Give the general partial fraction decomposition form

i.e $\dfrac{m}{(x^2+k)^2} \rightarrow \dfrac{Ax+B}{x^2+k} + \dfrac{Cx+D}{(x^2+k)^2}$

a) $\dfrac{x^4-2x^3+x^2+6x-5}{x^2-2x+1}$

$= \dfrac{x^2(x^2-2x+1)+6x-5}{x^2-2x+1}$

$= x^2 + \dfrac{6x-5}{x^2-2x+1} = x^2 + \dfrac{6x-5}{(x-1)^2}$

$\Rightarrow \boxed{x^2 + \dfrac{A}{x-1} + \dfrac{B}{(x-1)^2}}$

b) $\dfrac{x^2+x+1}{(x+4)(x^2+2)^2}$

$\Rightarrow \boxed{\dfrac{A}{x+4} + \dfrac{Bx+C}{x^2+2} + \dfrac{Dx+E}{(x^2+2)^2}}$

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2) Evaluate using partial fraction decomposition

a) $\int \frac{x^4 - 2x^3 + x^3 + 6x - 5}{x^2 - 2x + 1} \, dx$

$= \int \frac{x^2(x^2 - 2x + 1) + 6x - 5}{x^2 - 2x + 1} \, dx$

$= \int x^2 + \frac{6x - 5}{x^2 - 2x + 1} \, dx = \int x^2 \, dx + \int \frac{6x - 5}{(x-1)^2} \, dx$

$= \frac{x^3}{3} + c + \int \frac{6x - 5}{(x-1)^2} \, dx$

$\frac{6x - 5}{(x-1)^2} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2}$

$\Rightarrow$ Multiply both sides by $(x-1)^2$

$6x - 5 = A(x-1) + B$

$= Ax - A + B$

$\Rightarrow A = 6, \; B = 1$

$\int \frac{6x-5}{(x-1)^2} \, dx = \int \frac{6}{x-1} + \frac{1}{(x-1)^2} \, dx$

$= 6 \ln(x-1) + (2x-2)^{-1} \ln((x-1)^2) + C$

$$\boxed{\frac{x^3}{3} + 6 \ln(x-1) + (2x-2) \ln((x-1)^2) + C}$$

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2 b) $\int \frac{10}{(x-1)(x^2+4)} dx$

$\Rightarrow \frac{A}{x-1} + \frac{Bx+C}{(x^2+4)}$

$Ax^2+4A+Bx^2+Cx-Bx-C = 10$

$A+B = 0,$ $\quad B=-A$, $C=B$

$4A-C=10,$ $\quad C=-A \Rightarrow 5A=10, A=2, B=-2$

$C-B=0$ $\quad C=-2$

$\Rightarrow \int \frac{2}{x-1} + \frac{-2x-2}{x^2+4} dx$

$= 2\ln(x-1) - 2\int \frac{x+1}{x^2+4} dx$

$= 2\ln(x-1) - 2\int \frac{x}{x^2+4} dx - 2\int \frac{1}{x^2+4}$

$= \boxed{2\ln(x-1) - \ln(x^2+4) - \tan^{-1}(x/2) + C}$

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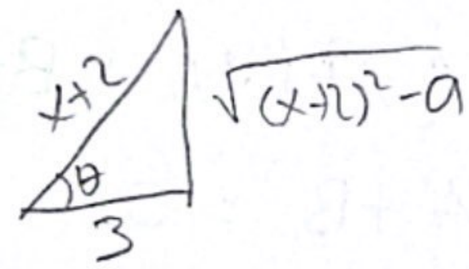
3) Evaluate $\displaystyle\int \frac{1}{(x^2+4x-5)(x+2)}\,dx$

$$= \int \frac{dx}{(x^2+4x+4-9)(x+2)} = \int \frac{dx}{((x+2)^2-9)(x+2)}$$

use $(x+2) = 3\sec\theta$

$x = 3\sec\theta + 2$

$dx = 3\cdot\sec\tan d\theta$



$$= \int \frac{1}{((3\sec\theta)^2-9)(3\sec\theta)}(3\sec\theta\tan\theta)\,d\theta$$

$$= \int \frac{3\sec\theta\tan\theta}{3\tan^2\theta\sec(\theta)}\,d\theta = \int \frac{1}{\tan\theta}\,d\theta = \int \cot\theta\,d\theta = \ln|\sin\theta| + C$$

$$= \boxed{\ln\left|\frac{\sqrt{(x+2)^2-9}}{x+2}\right| + C}$$

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4) Evaluate $\int x^2 \cos(3x)\, dx$

$u = x^2$ $\qquad$ $dv = \cos(3x)$

$du = 2x\, dx$ $\qquad$ $v = \frac{\sin(3x)}{3}$

$$\int x^2 \cos(3x) = x^2 \frac{\sin(3x)}{3} - \frac{2}{3}\int x \sin(3x)$$

$\int x \sin 3x$

$u = x$ $\qquad$ $dv = \sin(3x)$

$du = dx$ $\qquad$ $v = -\frac{\cos(3x)}{3}$

$$\int x \sin 3x = -\frac{x\cos(3x)}{3} - \frac{1}{3}\int \cos(3x)$$

$$= -\frac{x\cos 3x}{3} + \frac{\sin(3x)}{9} + c$$

$$\int x^2 \cos(3x) = \boxed{\frac{x^2 \sin 3x}{3} - \frac{2\sin(3x)}{27} + \frac{2x\cos(3x)}{9} + c}$$

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5 find the volume obtained by rotating the region bounded by $y=5x-x^2$ and $y=x$ about $x=4$.

$5x-x^2=x \Rightarrow x=0,4$

using cylindrical shells:

$$2\pi\int_0^4 (4-x)\left[5x-x^2-x\right]dx$$

$$=2\pi\int_0^4 16x-8x^2+x^3\,dx$$

$$=2\pi\left(8x^2-\frac{8}{3}x^3+\frac{x^4}{4}\right)\Big|_0^4$$

$$=2\pi\left(2(4^3)-\frac{8}{3}(4)^3+\frac{4^4}{4}\right)=\frac{128\pi}{3}$$

6 find the average value of $f(x)=\dfrac{1}{(x^2+4x-5)(x+2)}$

on the interval $[3,5]$

$$f_{ave}=\frac{1}{5-3}\int_3^5 \frac{1}{(x^2+4x-5)(x+2)}\,dx=$$

$$=\frac{1}{2}\left[\ln\left|\frac{\sqrt{(x+2)^2-9}}{x+2}\right|\right]_3^5$$

$$=\frac{1}{2}\left[\ln\left|\frac{2\sqrt{10}}{7}\right|-\ln\left|\frac{4}{5}\right|\right]$$

$$=\boxed{\frac{1}{2}\ln\left|\frac{5\sqrt{10}}{14}\right|}$$

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